AGEC 111:- INTRODUCTION TO AGRICULTURAL ECONOMICS

Definitions of Economics

- Economics is the study of how resources are used to satisfy the needs and desires of people.
- It is a science of choice making.
- It deals with allocation of scarce resources among competing alternatives.
- It is a science of production: resources are used to produce goods and resources for the satisfaction of human wants.
- It is the study of the use of the scarce resources for the achievement of alternative ends.
- It is a social science which covers the actions of individuals and groups of individuals in the processes of producing, exchanging, and consuming of goods and services.

Concepts of Economics

The followings are the basic concepts of economics;

1) Wants: human wants are many and insatiable. Everybody wants every good thing of life regardless of whether he or she could afford it or not. In other words, wants are wishes which people only dream to have without any serious efforts to actualize the dream.

2) Scarce resources: It is not possible to satisfy all wants because the means of satisfying them are limited or scarce. In other words, the available resources have alternative uses. They cannot satisfy all the alternative uses, they can only satisfy a limited number of uses hence they are regarded as scarce or limited.

3) Scale of preference: Since man's wants are many and the means of satisfying them are limited he has to draw a scale of preference by arranging his wants in order of priority and importance. In doing this, the most important ones are listed on the top of the table of preference while the least important ones are found on the lower rank of the table.

4) Choice making: Since man's wants are many and the means of satisfying them are limited, he makes a choice of the top most item in his scale of preference to satisfy his very important want as listed in his table of scale of preference while the other wants are left until resources are available to satisfy them.

5) Opportunity cost: This is the real cost of the item chosen on the man's scale of preference. In other words, the real cost of the item chosen on the scale of preference is the alternative want forgone, that is the next important item on the scale of preference that was not chosen due to scarcity of resources.

Economic Problems

Scarcity and alternatives are important ingredients for a problem to bear economic problem. For instance;

- **i.** If there is scarcity but no alternative, choice making is impossible and the problem is not economic in nature.
- **ii.** If there are alternatives but no scarcity, that is, goods are free, economics is not required.
- **iii.** Because resources are limited in supply, resource allocation is important regardless of the economic and political system of a country.
- **iv.** Products must be produced and resources must be allocated regardless of the political system – Capitalist, socialist and mixed states.
- **v.** The type of governments only take decisions on; what, how and when to produce and the allocation of resources for the production of the goods and services.

MEANING OF AGRICULTURAL ECONOMICS

Definition of Agricultural Economics

Agricultural economics is the application of economic principles to the operations of the agricultural industry. Like the pure economics, it is concerned with resource allocation in the agricultural industry.

Subject Matter of Agricultural Economics

The discipline of agricultural economics grew out of farm management. Farm management involves the application of scientific and technical principles to the solution of the day to day problems facing the farmers on the farms. The early farm management experts were mainly interested in the overall operation of the farm business and they did not specially relate the discipline to economics until a few agriculturists with training in economics transferred economic theory such as profitability, productivity and efficiency analyses into the discipline of farm management. For instance, the various disciplines of agriculture, such as, Animal science, Agronomy soil science, Agricultural Engineering, Agricultural Biology, Animal Health, agricultural biochemistry/Nutrition, all work towards producing more food and fibre for the population. There is always a tendency for experts in the different disciplines to carry out their assignments in artificial environment, that is, without minding the economic implications of their various actions and thus producing in an uneconomic stage of production, thereby leading to economic losses of scarce resources.

Therefore, the various disciplines in Agriculture have to learn to bring in economic analysis of the real farm situation where plants and animals have to be produced for a market with a number of socio-economic determinants influencing the resulting prices, costs and income. In agricultural economics, we study among other things, the following:

- Organization of farms.
- Availability of inputs and their prices.
- Marketing system for the various agricultural products.
- Organization of both production marketing cooperatives.
- Demands for and supply of agricultural products and those of allied industries.
- Finances of agriculture, role of banks, and other financial institutions in agricultural development.
- Peasant farmers to identify their problems as well as the problems of government owned farms.

Field of Agricultural Economics

Agricultural economics is divided into the following sub-disciplines or subject areas:-

i) Farm management
ii) Agricultural finance
iii) Agricultural marketing
iv) Agricultural resource economics
v) Agricultural extension education
vi) Agricultural project planning and appraisal
vii) Agricultural policy development
viii) Agricultural cooperation
ix) Agricultural business management.

DEMAND, SUPPLY AND PRICE THEORY

THEORY OF DEMAND

Meaning of Demand

Demand for a good or service is the quantity the consumer is prepared to buy of the good or service at a particular price. Demand is a desire backed up with the ability to pay for the good or service in question, thus it is not a wish.

There are two types of demand; individual demand and market demand.

Individual Demand

This is the schedule of amounts of a commodity which a person would buy at various possible alternative prices over a particular interval of time. Therefore, demand is a function of time and price. For example, a housewife's demand schedule for beef in a month is presented in Table

Individual Demand Schedule.

Market Demand

A market demand is a schedule of total amounts that would be purchased by all buyers in a particular commodity market at various possible alternative prices within a particular interval of time. It is the summation of individual demand schedules.

Examples of commodity market include; egg market, vegetable market, yam market, fish market and so on. The market demand of beef buyers is presented in Table 2.

A single individual demand cannot affect the market prices for a particular commodity because there are many buyers in the market, but changes in the market demand will affect the market, and thus it is the market demand that is important in economic analysis.

Individual and Market Demand Schedules.

Law of Demand

The quantity demanded of a commodity increases as the price decreases. In other words, the quantity demanded of a commodity decreases as the price of the commodity increases. Therefore, from the definition of demand, it follows that the prices and quantities demanded of a commodity are inversely related.

Mathematically, it is expressed as:

$\mathbf{Q_d} = \mathbf{f}\ (\mathbf{1/P})$.

Where: $\mathbf{Q_d}$ = Quantity demanded of the commodity, **P** = unit price of the commodity, **1/P** is an inverse function of price.

The above expression indicates that quantity demanded of a commodity ($\mathbf{Q_d}$) is an inverse function of the price of the commodity (**P**).

The law of demand is violated if:

(a) The commodity is a Giffen (inferior) good: Basic foodstuffs like potato, maize, yam and rice fall into this category. Most people do not vary their purchases as the prices of basic food stuffs increase (decrease) because the expenditure on them forms a small percentage of one's total outlay on food.

(b) There is fear of a future increase in price: The fear that the present increase in price is just the beginning of such increases in the future may not prevent people from buying more of during even though its price is increasing. For example, increasing prices of sugar during Christmas celebration may be seen as a sign that sugar would become scarce as the festival draws nearer, so people may rush to buy more with the increasing prices.

(c) The article is that of ostentation: Articles of ostentation, such as jewelries, expensive cars and clothes fall into this category. Rich women rush at expensive jewelries and clothes in place of the cheap ones while rich guys are crazy for state of the art expensive cars, with the assumption that the more expensive these goods are the more superior quality-wise.

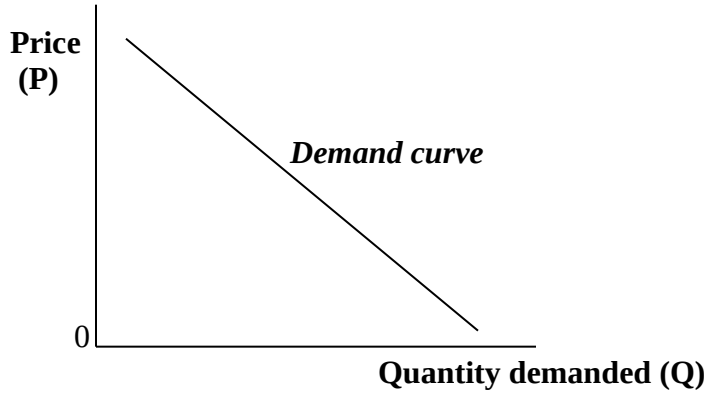
(d) The flooding of the commodity in the market.

Demand Curve

The demand curve shows the plotting of the prices on the vertical axis against the quantity demanded on the horizontal axis.

Figure 1 shows the demand curve for beef of the market demand schedules of Table 2 where quantity demanded is in kg and price is in shillings.

The demand curve slopes from left to right.



Slope of the Demand Curve

The slope of the demand curve is got mathematically from figure above thus:

$$\text{Slope} = \frac{0Q1 - 0Q2}{0P1 - 0P2}$$

The slope of the demand curve is negative because:

- Price is inversely related to quantity demanded.
- A fall in the price of a normal good makes it less expensive and more is purchased of such good (commodity). While a rise in the price of a normal good makes it more expensive and less of it is purchased.
- A fall in price raises the purchasing power or real income of consumers while a rise in price reduces the purchasing power of the consumer.

Factors Influencing the Demand for a Commodity:

The following factors affect the demand for a commodity, either in the form of a movement along the demand curve or real shift in the demand curve.

a) Price of the commodity: Demand for a commodity increases with decrease in the price of the commodity *ceteris paribus*. This is a negative relationship, that is, the quantity demanded of a commodity is inversely related to its prices *ceteris paribus.*

b) Price of Substitutes: An increase in the prices of substitute's results in an increase in the demand for the commodity while a decrease in the prices of substitutes leads to a decrease in the demand for the commodity. For example, a fall in the price of beef may lead to switching

from fish to beef. This is a positive relationship, that is, the quantity demanded of a commodity is directly related to the prices of its substitutes.

c) **Taste and fashion:** The demand for a particular commodity depends on taste and fashion. Fashion is apt to change more arbitrarily without reason. It is more fashionable for undergraduates and academicians to have laptop computers than desktop computers nowadays, hence the rush by this category of people for laptop computers.

d) **New commodities replace old ones (obsolete goods):** New inventions are constantly being made and the demand for these takes the place of the demand for the things they replace. For example, demand for desktop computers has replaced the demand for main frame computers which was in vogue in the 1970s.

e) **A change in real incomes:** A change in real income means a change in the quantity of goods and services the money income will buy. An increase in people's real income may have little effect on the demand for giffen goods (basic food stuffs) but it may have considerable effect on the demand for luxury goods. A university graduate that has just been employed by one of these new consolidated banks on a fat salary would be eyeing one of the posh cars in town or getting married the society-way than spending his income to increase his purchase of rice and meat.

f) **Population and its composition:** The more people we have to cater for, the more the demand for food items. The more children in the population, the more the demand for toys, baby dresses and educational facilities. Other variables under population that affect demand are wealth, credit availability, stock and habit.

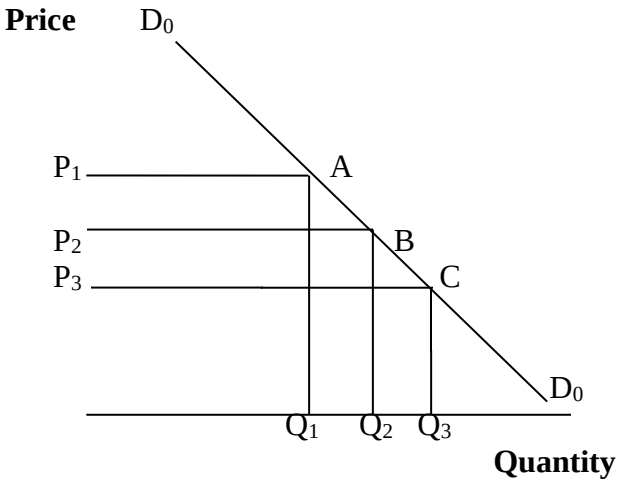
g) **A change in the quantity of money:** An increase in income will lead to increase in quantity demanded of a normal and luxury goods.

Shifts in Demand

All the factors mentioned above would affect the demand curve, either as mere movement along the demand curve or outright shift of demand curve upward or downward.

- Changes in the price of the commodity determine the slope of the demand curve, that is, they explain the movement along the demand curve.
- Changes in the other factors determine the position of the demand curve. An increase in the factors would shift the demand curve away from the origin while a decrease in the factors would shift the demand curve towards the origin.
- Factors which shift the demand curve inward or outward are called demand shifters.

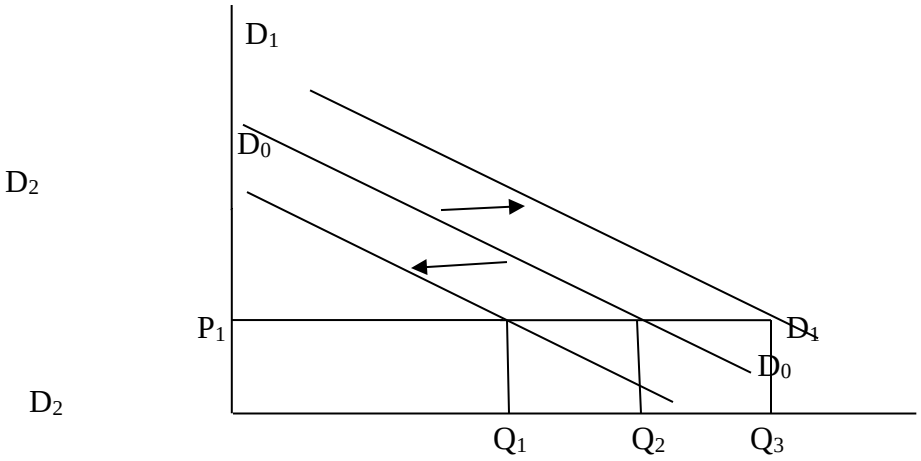
In figure below, movement from points A to B is still on the same demand curve. The movement was caused by changes in the price of the commodity. When the price of the commodity was P1 Quantity demanded was Q1 and the consumer was in point A. but when the price decreased to P2, quantity demanded was increased from Q1 to Q2 and the consumer was in point B.



Shifts in Demand

In the following figure, the three demand curves depict shifts in the demand.

The initial demand curve D0D0 shifted to D1D1 with decrease in factors such as income, price of substitute, population or fashion/taste, but with increase in the factors, the demand curve moves from D0D0 to D2D2.



THEORY OF SUPPLY

Meaning of Supply

This is the quantity of a commodity that is brought forth into the market over a particular period of time by a certain price.

Supply is not the entire stock (production) of any commodity in existence but only that amount drawn into the market by the price ruling at the time.

Supply depends on ruling price and scarcity. The supply of any commodity is limited because factors for its production have alternative uses. There are individual and market supply.

Individual Supply

This is the schedule of the amounts of any commodity which a seller would bring to the market at various possible alternative prices within a particular interval of time. The seller of the beef used in the demand example may have the following supply schedule.

Individual Supply Schedule

Price/Kg of beef	Kg of beef supplied/month
250	1.0
300.	2.0
350.	3.0
400.	4.0
450.	5.0

Market/Aggregate Supply

A market or aggregate supply is a schedule of total amounts that would be brought to the market by all sellers of the commodity at various possible alternative prices within a particular interval of time. It is the summation of individual supply schedules.

A single individual supply schedule cannot affect the market or the market price except in combination with other sellers. It is the market supply that is important in economic analysis.

Individual and Market Supply Schedules.

Price	A	B	C	D	E	F	G	H	Market supply
250	13	2.5	24	21.5					17
300	2	4.5	3.5	31.54	42.5				25
350	3	5	4	3	2	4		4	30
400	4	6	5	4	2	7	6	6	40
450	5	7	8	5	4	8	8	10	55

Law of Supply

The law states that the quantity supplied of a commodity increases as the price increases. It also implies that the quantity supplied decreases as the price decreases. In other words the prices and quantities supplied are directly related.

Mathematically expressed:

$\mathbf{Q_S} = \mathbf{f(P)}$.

Where,

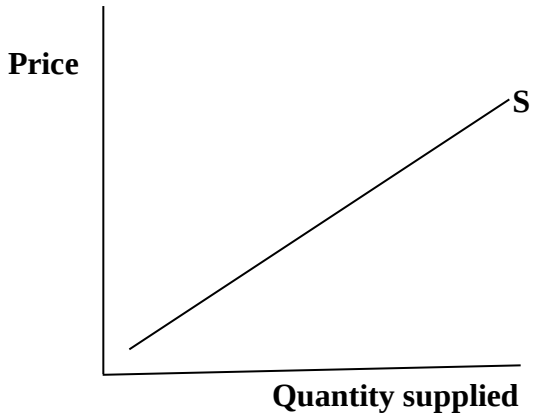
$\mathbf{Q_S}$ = Quantity of the commodity supplied to the market

P = Unit price of the commodity

The law of supply could be violated under these situations.

a) **Fixed Supply:** The supply of goods whose quantity is fixed and which cannot be increased as a result of economic decisions. The seller may decide not to even sell the commodity rather he may buy it himself. Examples: leisure, working overtime and supply of land.
b) **Infinitely inelastic supply:** Supply will not increase at all however much price rises. This happens when elasticity of supply is zero.
c) **In the case of labour supply:** Workers may wish to work shorter hours if wages (price of labour) are high.

Supply Curve
This is the plotting of the prices on the vertical axis against the quantities supplied on the horizontal axis.



The supply curve slopes upward from origin to the right. The slope of the supply curve is positive.

Factors Influencing the Supply of a Commodity
The under listed factors affect the supply of a commodity either as movement along the supply curve or outright shift.

(i) Price of the commodity: Supply increases with increase in price of the commodity *ceteris paribus.*

(ii) Cost of production of the commodity: Quantity supplied of a commodity is inversely related to the cost of production *ceteris paribus*. For example, if the cost of producing poultry increases due to high cost of feeds, vaccination and management, many poultry farmers would cut down on number of birds produced.

(iii) Availability of productive capacity: This is the capacity of a firm to produce more. This may not be feasible for a firm that is already producing at maximum capacity. Therefore quantity supplied is directly related to productive capacity.

(iv) Technological change: Improvement in technological change would lead to increase in supply

(v) Favourable climatic factors: Agricultural production depends directly on favorable climatic factors, such as; adequate and well distributed rainfall, enough sunshine and good relative humidity. A good weather condition would lead to higher agricultural production.

Movement along the Supply Curve

- The price of the commodity determines the shape of the supply curve. As prices change, the supplier changes the quantity supplied along the same supply curve. This is movement along the same supply curve.

Shifts in supply

- Changes in the other factors enumerated above determine the position of the supply curve. Any variable that reduces the cost of production of a commodity would result in increased supply of the commodity and the supply curve would shift to the right.
- Any variable that increases the cost of production of a commodity would result in decreased supply of the commodity and the supply curve would shift to the left.

CONCEPTS OF ELASTICITIES OF DEMAND AND SUPPLY

The concept of elasticity is to show the type of relationship existing between quantities of commodity and the factors that affect the demand and supply as both quantity and any of these factors change.

Meaning of Elasticity

Elasticity measures the responsiveness of quantity (demand or supply) to changes in the factors that affect the demand or supply.

Measurement of Elasticity

Elasticity could be measured as point or arc elasticity.

(a) Point elasticity: is the elasticity measurement if the changes in the variables of interest are very small, that is, at a point in both quantity and factors of interest. It is the first derivative of the demand or supply function.

(b) Arc elasticity: is the elasticity measurement if the changes in both quantity and the factors of interest are large.

Elasticity of Demand

Elasticity of demand is the responsiveness of quantity demanded of a commodity to changes in the factors that cause the change in the demand.

The important elasticities of demand are:

1. Price elasticity of demand
2. Income elasticity of demand
3. Cross elasticity of demand

1. Price Elasticity of demand

This is the measure of the responsiveness of demand to changes in the commodity's own price. The point price elasticity of demand is the first derivative of the demand function. It is differentiating the demand function with respect to price.

Price elasticity of demand is negative because the demand function (curve) has a negative slope.

Basic Determinants of Price Elasticity of Demand

a) Availability of substitutes: The demand for a commodity is more elastic if there are close substitutes for it.

b) Time period: Demand is more elastic in the long run.

c) **Number of uses to which a commodity can be put.** The more the possible uses of a commodity the greater its price elasticity will be.
d) **The nature of the need that the commodity satisfies.** Luxury goods are price elastic while necessities are price inelastic.
e) **The proportion of income spent on the particular commodity.**

Types of Price Elasticity of Demand
There are five types of price elasticity of demand.

(i) Perfectly Elastic Demand

A small decrease in price gives rise to an indefinitely large increase in amount demanded. The demand curve for a perfectly elastic demand is given in Figure 7. The demand curve is a horizontal line and the numerical elasticity of demand is infinite, that is, $Ep = \infty$

(ii) Perfectly Inelastic Demand

In a perfectly inelastic demand, whatever the original price is, a reduction in that price will lead to no increase at all in the amounts demanded. The demand curve is a vertical line and the numerical elasticity of demand is zero.

(iii) Unitary Elasticity Of Demand

A proportionate change in the price of a commodity results in the same proportionate change in quantity demanded. The curve slopes from left to right. The numerical elasticity of demand under the unitary elasticity is equal to one throughout the length of the demand curve.

(iv) Elastic Demand:

A proportionate change in the price of a commodity results in more than proportionate change in quantity demanded. The numerical elasticity of demand is between one and infinity, that is,
$1 < Ep < \infty$

(v) Inelastic Demand:

A proportionate change in the price of a commodity results in a less than proportionate change in quantity demanded. The numerical elasticity of demand is between zero and unit, that is,
$0 < Ep < 1$

Uses of Price Elasticity of Demand
It is used as a planning tool. To classify goods as superior and inferior goods;

- Luxuries have high elasticity of demand. Examples of luxuries are jewelries, expensive cars and so on.
- Necessities (normal goods) have small elasticity of demand. Example is basic food stuffs. Yam meat, rice etc. they have elastic demands and the numerical elasticity is between unit and infinity.
- Giffen goods have inelastic demand and the numerical elasticity is between zero and unit.
- Salt has perfectly inelastic demand and the numerical elasticity is zero.

2) Income Elasticity of Demand

It is the proportionate change in the quantity demanded of a commodity resulting from a proportionate change in the income of the consumer.

Income elasticity can be positive or negative.

Basic Determinants of Income Elasticity of Demand

- The nature of the need that the commodity covers. The percentage of income spent on food declines as income increases (this is Engel's law).
- The initial level of income of a country. A personal computer is a luxury in the developing country, whereas it is a necessity in the developed country with high per capita income.
- The time period, because consumption patterns adjust with a time lagto changes in income.

Uses of Income Elasticity on Demand

Income elasticity of demand is used to classify goods into luxuries, necessities and inferior goods.

- Luxuries have income elasticity that is greater than unity, that is, Ey> 1
- Necessities have income elasticity that is between zero and unity, that is, 0 <Ey< 1. .
- Inferior goods have income elasticity that is less than zero, that is, Ey< 0.

NB/A good may not be superior or inferior every time, it depends on the income level of the consumers (Engel's law).

3) Cross Elasticity of Demand

It is the proportionate change in the quantity demanded of a commodity(X) resulting from a proportionate change in the price of another commodity (Y).

The basic determinant of cross elasticity is the nature of the commodities relative to their uses. If two commodities can satisfy equally well the same needs, the cross elasticity is high, if not the cross elasticity is low.

Uses of Cross Elasticity

It is used to classify goods as complementary and substitutes goods.

- Complementary goods X and Y have negative cross elasticity
- Substitutes have positive cross elasticity
- The higher the value of the cross elasticity, the stronger is the degree of substitutability or complementarity of commodities X and Y.

Elasticity of Supply

It is the measure of the responsiveness of supply to changes in the commodity's own price.

The price elasticity of supply is positive.

Types of Elasticity of Supply

- Unitary elasticity of supply.
- Zero elasticity of supply.
- Infinite elasticity of supply.

Demand and Supply Analysis in Agriculture

A. Agricultural products are inelastic in demand because they aregenerally necessary for human existence.
B. Agricultural products are inelastic in supply because of theseasonality in their production and thus, their production cannoteasily be varied no matter the demand.

C. The inter-action of demand and supply determines the market pricesof agricultural products.
D. In the events of decreased demand during the peak season and toencourage increase production, government can give price support orguaranteed minimum prices.
E. To decrease the cost of producing a commodity, government cansubsidize the price of the resources used in the production of thecommodity. Such resources or inputs include fertilizer, seeds, agrochemicals, machinery and equipment.
F. Price controls of agricultural products during the period of shortagesto protect consumers from paying exorbitant prices.

PRICE THEORY

Meaning of Price

The price of a commodity is the rate at which the commodity can beexchanged for anything else.
a) Price is a signal to producers to expand or contract their rate ofproduction.
b) It reflects the marginal social value of a commodity.
c) It is used to ration goods among prospective buyers in the market.

Price determination

Prices are determined by the following processes.
(a) Haggling. This is the process of bargaining on the price of thecommodity between a few sellers and a few buyers. The seller says aprice probably higher than the price he expects the good to be sold.The buyer then presents a lower price and after a period ofbargaining a final price is reached. This is how price is mostlydetermined in Kenyan local markets.
(b) Sales by Auction: This involves one seller and a number ofprospective buyers. The good is offered for sale, the buyer bid for itand the good is sold to the highest bidder, that is, the one prepared topay the highest price. In sales by auction, the cost of production oracquisition cost does not enter into the determination of the price.
The price of the good depends on the relation between the keennessof buyers to buy and the keenness of sellers to sell.
(c) Offers at fixed prices: In the advanced countries and evensupermarkets in Kenya, prices of; foodstuffs, clothing, andhousehold goods and so on are pre-determined at definite prices. Theprices are marked and pasted on the goods. The seller fixes his pricesafter due considerations for his acquisition and marketing costs.
(d) Equilibrium price: The market price is determined by the interplay ofthe forces of demand and supply. The equilibrium price is the pricethat clears the market, that is, the price that clears the amount ofcommodity supplied and demanded or the price when quantitysupplied equals quantity demanded. The quantity is also calledequilibrium quantity.

Market and Normal Prices

(a) Market price. This is the price charged in the market in the short runperiod, that is, the short term equilibrium price when supply and demand may be subject to temporary influences.
(b) Normal price: This is the price charged in the market in the long runperiod when the conditions of supply and demand have stabilized and the rate of consumption of a good is equal to the rate at which itis produced. Normal price is the long run equilibrium price.

Stable Equilibrium Analysis

Stable equilibrium: The equilibrium is stable if demand and supply conditions are such that a displacement of the equilibrium situation automatically causes a return to it.

- Equilibrium will be stable when slightly to the left of the equilibrium position, the demand price exceeds the supply price and slightly tothe right of it the supply price exceeds the demand price.
- Equilibrium is also stable if at a price slightly above the equilibrium level, the amount demanded falls short of the amount supplied.
- The stability condition will always be fulfilled if the demand curves slopes downward and the supply curve slopes upward.

THEORY OF PRODUCTION

Introduction

Production is synonymous with creating something. Economist does not restrict production to the manufacture of commodities; he also takes the expression to include the provision of services such as those of lawyer, accountant, actor, musician, or market women. The central aim of all production is to satisfy people's wants and increase the economic welfare of a people, to raise their standard of living by enabling to satisfy more fully a greater number of their wants.

Definition of Production

Production is the process by which inputs are transformed into output. The output producing units include farms and factories.

Production therefore means changing either the form of something or its situation in space or time, or the provision of a service of some kind.

Agricultural Resources (Inputs)

The major resources used in agriculture are free and economic resources.

- The free resources are termed free because they are relatively abundant in supply and hence have no cost elements attached to them but they are highly essential in agricultural production.They are the air, heat, water (rain) and so on.
- The economic resources are scarce in supply and limiting in production. They possess high economic value. They are land,labour, capital and management

The economic resources come into agricultural production in different disaggregated forms. To give a more specific set of resources for example, land of different types, different categories of capital expenditure and labour of different categories such as hired labour, family labour,skilled or casual or unskilled labour.

Economic Agricultural Resources

Land: Land is the most important resource in agricultural production.

Its supply is relatively fixed except it can be expanded slightly by drainage of swamps, reclamation from sea bodies and water, andchemical or biological improvement of non-cultivable lands. Land (farmsize) is measured in hectares

Labour: Labour is the effort of human being that is used in makingthings happen in the production process. It is the second most importantresource next to land in agricultural production. Labour availability is afunction of the economically active proportion of the populationreleased for agricultural activities. Farm labour on peasant farms couldbe from both family or hired sources

depending on the size of the farmand type of operation to be performed, but in large commercial farms,farm labour is purely hired and categorized as casual, unskilled,semiskilled and skilled labour. The quality of labour in use is a functionof the level of education and training in relevant agricultural production,that is, the higher the level of training in the cultivation and managementof any crop the higher the productivity and efficiency.

Labour is measured in terms of the adult male equivalents, where oneman day is the work done by one adult male in eight hours.

Capital: Capital resources come into farm production in three forms,namely,

1) Farm machinery, such as, tractors and various farm tools
2) Biological capital such as fertilizers, pesticides, herbicides, improvedseeds and breeding stock
3) Feed for Livestock.

Management: Management or entrepreneur or co-ordination is the mostimportant factor of production. In this input resides the decision makingpower in farm business. It is concerned with efficient mixing ofresources in the production process. An efficient resources planning and utilization ensures attainment of the objective of the production function. Management is therefore concerned with planning,implementation and control of the farm business.

PRODUCTION FUNCTION

Definition of Production Function

Production function is a physical or technical relationship between inputs and output in any given production processes. It describes the rate at which inputs are transformed into outputs. It defines the production possibilities open to the farmers. In an implicit form, the productionfunction is defined as:

$Y = f(x)$.

It states that output Y is a function of input X.

Where Y = output of product, X = input used,f ( ) = functional form.

Types of Production Function

Production functions are classified into two based on time period. Theyare short run and long run production functions.

(a) Short run production function:

In the short run production function, at least one of the inputs used can be varied with the others fixed.

The implicit functional form is defined as:

$Y = f(X1, X2/ X3, X4\ldots Xn)$.

Or

$Y = f(X1, X2)$.

It implies that output (Y) is a function of variable inputs X1 and X2 given fixed values of inputs X3 to Xn.

(b) Long run production function:

In the long run production function, all the inputs can be varied.

It is implicitly expressed as;

$Y= f(X1, X2, X3\ldots Xn)$.

Uses of Agricultural Production Function

1) They serve diagnostics purposes to diagnose input and output relationship within sampled farms.
2) They tell us the level of optimum use of inputs.
3) Determination of the scale of production.
4) They are useful in determination of enterprise combination.
5) They guide farmers in decision making.

Types of Functional Forms

The common algebraic forms of production function are the linear, quadratic, exponential and power or Cobb Douglas

Production Function Parameters of Interest

The following productivity measurements are derivable from the analysis of the production function. They include:

(a) Average Product

The average product (AP) measures the average rate at which an input is transformed into a product. It is a measure of technical efficiency of the variable inputs used in the production process. The AP is the total product (TP) divided by the input quantity (X).

If $Y = f(X)$.

$AP = Y / X = f(X) / X$

(b) Marginal Product

The marginal product (MP) measures the amount that total output increases as input increases. It is the change in output resulting from a unit change in the variable input. The MP is the slope of the production function curve. Mathematically it is derived thus:

If $Y = f(X)$

$MP = dY / dX$

(c) Elasticity of Production

Elasticity of production (Ep) is the proportionate change in output caused by a proportionate change in the variable input.

Ep = marginal product / average product.

(d) Values of Elasticity of Production

- The values of the Ep are between minus infinity and infinity, thatis, $-\infty < Ep < \infty$
- When $MP > AP$, $Ep > 1$
- When $MP < AP$, $Ep < 1$
- When $MP = AP$, $Ep = 1$
- When $MP = 0$, $AP > 0$, $Ep = 0$
- When $MP < 0$, $Ep < 0$.

(e) Returns to Scale

The return to scale (RTS) is the summation of the elasticity ofproduction of variable inputs involved in the production function, that is,$RTS = \sum Epi$

It is used to determine the stage the overall production process is in the production surface. There are three stages in the production function,namely;
(i) When RTS > 1, it implies production is in stage 1, the stage of increasing returns to scale. Production at this stage should be intensified by increased use of variable resources with positive Ep.
(ii) When 0 < RTS < 1, it implies production is in stage 11. The stage of economic efficiency or positive decreasing returns to scale.
This is the stage where every producer thrives to be to maximize profit, output and minimize cost.
(iii) When RTS < 0, it implies production is in stage 111. This is the stage of negative decreasing returns to scale or stage of economic inefficiency. This is the stage where there should be no production because with each additional use of variable input, overall total production decreases.

Types of Marginal Returns

Marginal return is the rate at which output is increased by each additional increase in input possibilities. There are three types of possibilities between input and output in the production function curve,they are: constant, increasing and decreasing marginal returns.

(a) Constant Marginal Returns (CRM)

The constant marginal returns states that each additional unit in inputresults in a constant rate of increase in output. The change in output isconstant throughout. The slope of the curve is equal throughout the length of the straight line curve.

(b) Increasing Marginal Returns (IMR).

In the increasing marginal returns, each additional increase in the inputsleads to more than increase in output than the preceding increase inoutput. The change in output increases at an increasing rate. The slopeof the curve becomes steeper as more inputs are added.

(c) Decreasing (Diminishing) Marginal Returns

In the decreasing marginal returns, each additional increase in the inputsleads to smaller increase in output than the preceding increase in output.
The slope of the curve becomes smaller as more inputs are added. Thechange in output increases at a decreasing rate until it becomes zero andeven negative. Decreasing marginal returns is common in mostagricultural production.

PRODUCTION RELATIONSHIPS: FACTOR –PRODUCT RELATIONSHIP

Meaning of Production Relationship

Production relationship describes the form of association between inputsinvolved in production and the product produced. The relationshipspecifies a production unit which makes use of input mix in obtaining agiven output.
There are three major types of production relationships between inputsand output, namely;

- Factor-product relationship
- Factor-factor relationship
- Product-product relationship

Factor Product Relationship

It describes the transformation of a given input into a product. There is only one product such as maize. We are interested in the effect of variation of only one input such as labour or fertilizer while other inputs are fixed. Many farm decisions are mostly analyzed using this production relationship, especially when the problem is to determine the intensity with which the given

variable input shall be combined with fixed quantities of other essential inputs to achieve the stated objectives.

The factor product relationship is not true to life because production always involves more than one variable input in the short run and all inputs are variable in the long run. It is only used for diagnostic purpose where one basic input is the limiting input e.g. labour or capital in developing economy.

The production function under the factor product relationship is of the implicit form;

$Y = f(X1/ X2 \ldots Xn)$, or $Y = f(X1)$

Where Y = output in specified unit, X1= variable input, X2 –Xn are fixed quantities of other inputs.

The total product (TP) = Y. $Y = f(X1)$.

The average product (AP)

$AP = Y/X1 = f(X)/X1$

The marginal product (MP)

$MP = dY/dX1 = f\,1(X1)$.

The Generalized Production Functions under Factor-Product Relationship

The total product (TP), average product (AP) and marginal product

(MP) are presented in table and figure below.

The analysis of the production function curve with the knowledge of the important production function parameters is used as a frame of reference to undertake a quantitative interpretation of the three stages of the production surface.

Stages of the Production Function Curve

There are three stages of the production function.

(a) Stage I

- It is a stage where the marginal product, MP, is greater than the average product, AP.
- Both MP and AP are increasing.
- MP is maximum at point of inflexion where the first derivative ofthe production function, dY/dX1, is zero and the second level derivation, d2Y/dX21 is also zero.
- MP equals AP at the boundary of stages 1 and 11 where theelasticity of production equals unit.
- Elasticity of production is greater than or equal to unit.
- It is a stage of increasing marginal returns.
- The economic decision is that if the product being produced has any valve at all, Input use and output production oncestarted should be continued until stage 11 is reached.

(b) Stage II

- It is a stage where the MP is less than AP.
- AP is still increasing.
- MP is decreasing but positive.
- MP is zero at the boundary of stages 2 and 3.
- It is a stage of economic relevance in production where total product increasing at a decreasing rate.
- It is a stage of decreasing positive marginal returns anddecreasing returns to scale.
- The elasticity of production is between zero and the unit.
- It is a stage of rational production where output and profit are maximized and input use is optimized.

(c) Stage III

In this stage;

- AP is greater than MP.
- AP is positive and greater than zero.
- MP is less than zero and negative.
- Both MP and AP are falling.
- Production is not advisable in this stage because increase in input use leads to reduction in total product.
- Elasticity of production is less than zero that is, it is negative.

Optimal Input Combination

Economic efficiency in factor product relationship is attained when thenecessary and sufficient conditions are met. This implies integrating thephysical concept (necessary condition) with the economic principles(sufficient condition).

The necessary condition for optimum production is when the marginalproduct is equal to zero. That is, Total Product is maximum when MP =0.

For the sufficient condition the profit maximizing combination of inputsis given by equating the marginal product of each input with the priceratios of the variable input and product, that is,

MP = Px / Py.

Where,

MP = marginal product of variable input X

Px = unit price of X

Py = unit price of product Y

A further analysis of the above condition gives more conditions forreaching optimum or maximum profit, that is, profit is maximizedwhen,

MP = Px / Py

Or

Py(MP) = Px.

That is when the value of marginal product MVP, (PyMP) equals theunit price of the input.

PRODUCTION RELATIONSHIPS: FACTOR-FACTORRELATIONSHIP

Meaning of Factor-Factor Relationship

This is a situation whereby two variable inputs are used in theproduction of one product. Each combination of the inputs produces aunique amount of output. In this relationship, changes in the product (Y)are due to possibilities of substituting one input for the other input, that is, substituting X1 for X2 or X2 for X1 as Y is held constant at a particularlevel.

Implicitly, a factor – factor production function is defined as:Y = f (X1, X2 / X3… Xn),

OrY = f (X1, X2)

Where Y is the output, X1 and X2 are the variable inputs with inputs X3to Xn fixed.

In factor – factor relationship, a given level of output can be producedwith different combinations of inputs

Important Economic Parameters of Factor-FactorRelationship

Isoquant

This is the curve of all combinations of X1 and X2 which yield specifiedoutput levels. It is also called iso-product or equal product curves. Amovement on each isoquant implies the same output level but withdifferent combinations of X1 and X2.

Properties of Isoquants:

i. Two isoquants will never cross because each combination ofinputs can only produce one and only one amount of output.

ii. Isoquants are either convex to the origin or slopes from left toright.

iii. A convex isoquant implies diminishing marginal production andproduction is in stage 11 or stage of rational production orefficient stage of production.

iv. Isoquant has negative slope.

Marginal Rate of Input Substitution (MRIS)

MRIS is the rate at which inputs substitute for each other to maintain aconstant output. It is obtained from a total differentiation of theproduction function:

$Y = f(X1, X2)$

$dY = f1dX1 + f2dX2 = 0.$

$f1dX1 = - f2dX2$

$f1/ f2 = - dX2/dX1 = MRIS(x1 \text{ for } x2).$

Where, f1and f2 are the partial derivatives of Y with respect to X1 and X2respectively. The f1 and f2 are also the marginal products with respect toX1 and X2 respectively.

(i) MRIS is the negative of the ratio of two marginal products. Thatis, $MRIS = - f1/f2$.

Isocline

It is a line which connects points of equal slope on a production surface.

That is, it passes through points of equal MRIS on an isoquant map. On the isocline for each input $MRIS = k$.

Expansion path

This is the line joining points of least cost combination of inputs on the production surface. These are points where isocost lines are tangent tothe isoquants. On the expansion path for each input, the marginal rate ofinput substitution of input X1 for X2 equals the ratio of the input prices, that is, $MRISX1X2 = Px1 / Px2$.

Where, Px1 and Px2 are unit prices of inputs X1 and X2 respectively.

Optimal Inputs Combination and Profit Maximizationunder Factor – Factor Relationship

Economic efficiency in factor – factor relationship is attained when thephysical concept (necessary condition) is integrated with the economicprinciples (sufficient condition).

The necessary condition is met when marginal rate of inputsubstitution is zero, i.e., $MRIS = 0$.

The sufficient condition is met by employing the least cost criterion.

Least cost criterion

The least cost criterion is met by equating the MRIS with the negative ofthe price ratio of inputs, that is, the least cost combination is when;

$MRIS = - Px1/ Px2$

Graphically, the least cost combination of inputs occurs at the pointwhere the isocost line is tangent to the isoquant given that the isoquantis convex to the origin.

Isocost Line

The isocost line is the curve of all combinations of two inputs that costthe same amount, that is,

PRODUCTION RELATIONSHIPS: PRODUCT - PRODUCTRELATIONSHIP

Meaning of Product-Product Relationship

This is a production relationship involving the production of twodifferent enterprises such as wheat and potatoes with only one variableinput such as land or labour. It involves substitution between the twoenterprises given the variable input. In other words, to produce morehectares of wheat with a given hectares of land will imply theproduction of less hectares of potatoes and vice versa. Product – productrelationship determines the combination of enterprises that best meetsmanagement objectives given the resource limitation.

Implicitly, the production function is of the form;

X = (Y1, Y2)

Where X = variable input.

Y1 = enterprise 1 (wheat).

Y2 = enterprises 2 (potatoes).

Production Possibility Curve (PPC)

This is the curve of the combination of two enterprises that can beproduced with a given set of input. It is to determine the most profitablecombination of enterprises for a limited amount of input. The PPC isalso called Iso-resource curve because each point on the curverepresents combinations of outputs from equal amounts of input.

Marginal Rate of Product Substitution (MRPS)

This is the rate at which one product is substituted for the production ofthe other product, given a limited level of variable input use. It isobtained from a total differentiation of the product possibility function.

X = h(Y1, Y2) which states that the cost of production in terms of X is afunction of the two products Y1 and Y2.

Relationship among Products: Types of Products

An examination of the marginal rate of product substitution allows us toclassify products into competitive, complementary, supplementary andjoint products.

Competitive Products

For competitive products the MPRS is less than zero. This implies thatoutput of one product can only be increased by reducing the output ofthe other product. Competitive products require the same set of inputsand one product can be expanded by diverting inputs to its productionand thus reduce the quantity of inputs to other. Examples of competitiveproducts are wheat and oat, oil palm and cocoa, cassava and yam,yam and cocoyam, maize and rice and so on.

Complementary Products

For the complementary products, the MPRS is greater than zero. Thisimplies that increasing the production of one by diverting the use ofmore inputs to its production leads to increase in the production of theother product. This is very common in agriculture.

Examples of complementary products are;

Legumes and grass crops in crop rotation.

Beef cows and hog (pigs) produced together. The pigs feed on thewastes of the beef cows.

Supplementary Products

In supplementary products the MPRS is zero. Two enterprises bearcomplementary relationship to each other when the output of oneproduct can be increased without reduction or increase in the output ofthe other product. Supplementary products are common in agriculture. For example, use of family labour for gardening and poultry enterprise. Theproduction of up-land and swamp rice using labour and machinery,Swamp rice does not need machinery for production; therefore, usingmachinery for upland rice does not affect the production of swamp rice.

Joint Products

Joint products are always produced together in the same productionperiod. Examples are palm oil and palm kernel. Beef and hides and soon.

Optimal Product Combination and Profit Maximization under Product-Product Relationship

Economic efficiency in product- product relationship is attained whenthe following necessary and sufficient conditions are met.

(a) Necessary condition:

MPRS = 0.

(b) Sufficient condition:

When the MPRS is equated with the negative ratio of the pricesof products Y1 and Y2, that is,

MPRSY1→Y2 = -Py1 / Py2

Where Py1 and Py2 are unit prices of products Y1 and Y2respectively

THEORY OF COST

Meaning of Cost

Costs from the perspective of the farm are the:

i. Expenses incurred by the farmer on the operations of his farmbusiness.

ii. Values in monetary terms of the inputs used in production of theproduct of the farm.

iii. Outlay of funds on goods and services used in organizing andcarrying out the production process.

MEASUREMENT OF COST

Measures of cost

Classical Measurement of Cost

Under the classical measures of cost, there are three major componentsof cost. They are namely: total cost, variable cost, fixed cost.

(a) Fixed Costs

These are the costs of fixed inputs of production. They do not vary withthe level of output. They may be incurred even when production is notundertaken as is the case with building a factory or farm house orprocuring a tractor for farm business. For example, once a firm hasprovided itself with a factory and installed the necessary machinery, thecosts on these items remain the same whether the firm is working at fullor only half capacity. Examples of fixed costs include: salaries ofadministrative staff, expenses for land maintenance and rent, allowancesfor depreciation (wear and tear) of machinery, expenses for buildingdepreciation, interest on loans, charges for management and so on.

(b) Variable Costs

Variable costs are the cost of variable inputs used in the productionprocess. Variable costs vary with the level of output and are thus afunction of output. This is a category of cost that varies in the short runperiod while in the medium run and long run, all costs become variable.
For example administrative costs become variable in the medium period.
Examples of variable costs are wages of labour, cost of rawmaterials (feeds, seeds, fertilizer, Agro chemicals etc), electricity bills,and cost of powering the generators and other repairs and maintenanceto plant and machinery and factories.
Mathematically the variable cost is a linear function of this form:
TVC = Px1X1 + Px2X2.
Where TVC = total variable cost.
Px1 = unit price of variable input X1.
Px2 = unit price of variable input X2
X1 = variable input X1
X2 = variable input X2.

(c) Total Cost
Total cost is the sum of the variable costs and fixed costs.
It is expressed as:
TC = TVC + TFC
Where TC = total cost, TVC = total variable costs andTFC = total fixed costs.

AGRICULTURAL COST FUNCTIONS
Cost Functions
A cost function is a mathematical relationship between the total cost ofproduction and the output produced. It is derived from the productionfunction. For a factor – product production function of the form.
Y = f(X).
Where Y = output, Y is a function variable of the input X. It can also be written as;Y = f (Px, X), that is, output Y is a function of cost of variable cost ofvariable input X where Px is the unit price of variable input X and PxX isthe cost outlay on variable input X at various levels of X (assuming theunit price of X does not vary with the quantity of input purchased). Thecost function is therefore written as Y = f (PxX).

Shape of the Cost Functions
(a) Total Cost Function
Since the cost function is derived from the production function, theshape of the total cost curve is similar to that of the underlyingproduction function.
A generalized total cost function is the one for a production function thathas distinctly the three stages of production- constant, increasing anddecreasing returns to scale as explained under the analysis of theproduction function generalized cost function looks like an inverted letter.
(b) Total Fixed Cost Function
The shape of the fixed cost function is straight line that is parallel to theoutput axis because it does not vary with output leveland thus does not change during the short run period of production.
(c) Total Variable Cost
The total variable cost function looks like the total cost function.
It is different from the total cost function:

- It is below the total cost curve
- It starts from the origin while the total cost has an intercept equalto the size of the total fixed cost.

Analysis of Cost Functions

The cost functions when subjected to further analysis would result in thederivation of the average cost functions and the marginal cost functions.

These two cost metrics, that is, the average cost (AC) and marginal costs
(MC) are used in further economic analysis of the cost and profitfunctions of the farm business.

Average Cost

The average cost is the average cost of producing each unit of output.

Three categories of average costs are: Average total cost (ATC),
Average fixed cost (AFC) and Average variable cost (AVC).

(a) Average Total Cost (ATC)

This is got by dividing the total cost (TC) by the total product, (Y).

It is mathematically defined as;

ATC = TC / Y.

Where TC = Total cost and Y = Total Product.

The curve of the ATC is U shaped. It declines initially; it reaches aminimum at the level of optimal plant size and rises

(b) Average Variable Cost (AVC)

It is the total variable cost divided by the total product.

It is mathematically defined as;

AVC = TVC/Y.

Like the ATC:

The curve of the AVC is U-shaped.

It declines initially. It reaches a minimum and rises again

Marginal Cost

The marginal cost (MC) is the change in total cost (TC) due to unitchange in output (Y). It is the cost of producing an additional unit ofoutput. It is got by dividing the change in total cost by the change intotal product, that is,$\Delta TC / \Delta Y$.

Since the fixed cost does not change, therefore, the only change in totalcost, ΔTC is the change in total variable cost (ΔTVC).

The marginal cost is also defined as $MC = \Delta TVC / \Delta Y$.

The exact marginal cost is the first derivative of the Total cost function,that is, $MC = dTC / dY$.

Graphically,

- The MC is the slope of the TC curve
- It is also the slope of the TVC curve.
- The MC is also U shaped for TC and TVC curves that areinverted S –shaped.

(a) Relationship among MC, ATC, AVC and AFC

- MC equals ATC and AVC at their minimum point.
- MC equals ATC at an output greater than the output at whichAVC is at minimum because of the presence of some fixed costin the total cost.
- AFC slopes downward from left to right and asymptoticallyapproach the output axis as output increases.

- When MC < 0, MC < ATC.
- When MC > 0, MC > ATC.
- When MC = 0, MC = ATC.
- The slope of ATC becomes zero at the minimum point of ATCand MC equals ATC at the minimum point of ATC.

AGRICULTURAL MARKETING

Definition of Market

i. Market refers to any group of transactions or business dealingsbetween buyers and sellers.

ii. As opposed to simple selling, a market implies trade that istransacted with some regularity and regulation and in which acertain amount of competition is involved. This competitionarises whenever groups of buyers and sellers come together andallows the operation of supply and demand.

iii. And market can denote potential or estimated consumer demand.

iv. The earliest markets in history conducted bartering. After theintroduction of money, commercial codes were developed thatultimately led to modern national and international enterprises.

As production expanded, communications and middlemen cameto play an ever growing role in markets.

v. The term market is also used to denote: A place where peoplemeet to buy and sell goods or commodities on daily, every fivedays and even on weekly basis. This type of market is found inthe rural and urban areas with a basic feature of buyers andsellers meeting to exchange the title-ownership of thecommodities brought to the market.

vi. A market is any organization whereby the buyers and sellers of acommodity are kept in close touch with each other for theexchange of title-ownership of the commodity in the market. Thisparticular definition of market takes care of marketing activitiesusing the Internet facilities, Mobile Phones, and postal Services.

The buyers and sellers do not have to come to a particular placeas is the case with the rural and urban markets for their marketingactivities. They could contact themselves using the phones,internet facilities and the postal services to transact marketingactivities. The development of communications such as means oftransport, postal, telephone, internet facilities, telegraph services,banking and scientific grading of commodities have all helped towiden markets.

vii. Another way of defining market is to indicate the type ofcommodity and the extent of the sale of the commodity. This typeof market is called commodity market. Examples are coffee market, tea auctioning market, egg market and so on.

Conditions for a Market to Exist

The existence of any forms of a market depends on the following basicfactors, namely;

- The commodities to sell must exist
- There must be sellers who are ready to sell the commodities
- There must be buyers who want to buy the commodities put upfor sale
- The buyers must be able to strike bargains with the sellers on theprices of the commodities
- The process of striking a bargain is done through the process ofhaggling in most rural and urban markets. In most commoditymarkets, price determination is not by haggling but through aprocess of attaching specific prices to different quality grades ofthe commodity.

Importance of Markets

- ✓ The constitution of markets and market prices are central problemsof economics.

- ✓ The economic importance of a market is that it is a place whereprices of commodities are determined and title-ownership isexchanged.

TYPES OF MARKETS

Criteria for Categorizing Markets

There are three criteria for categorizing markets. They include;

i. Degree of Competition

ii. Volume of trade

iii. Commodity traded

Degree of Competition

Competitions in economics are conditions that are present in marketswhere buyers and sellers interact to establish prices and exchange goodsand services. Economic competition is the means whereby the selfinterestof buyers and sellers acts to serve the needs of society as well asthose of individual market participants.

Society is served when the maximum number of goods is produced atthe least possible prices.

Under the degree of competition, there are two main types of markets;

- ✓ Perfect markets
- ✓ Imperfect markets

Perfect Markets

The essential features of a perfect market are;

(a) Homogeneous Commodity

Any one unit of the commodity brought to the market is exactly like anyother. There is no trade name or trade mark to distinguish the supply ofone seller from that of another. It is of no consequence then from whichseller a buyer makes his purchases.

For example, if tomatoes are offered for sale, all the sellers will be selling tomatoes. If it is rice, all the sellers would be selling rice.

(b) Many Buyers and Sellers

A large number of buyers and sellers would make it impossible for afew buyers and sellers to significantly influence the market demand,supply and prices either by curtailing purchases and supply respectively.

Apart from being many, the many buyers and sellers must be on theaverage small firms. The significance of a large number of small firmsand many buyers is that the power to influence the behaviour of theparticipants in the market is thoroughly dispersed. In other words, nosingle person or business has the power to dictate the terms on which theexchange of goods and services takes place (unlike the situation in amonopoly or oligopoly). Market results then are truly impersonal.

(c) Buyers and Sellers must be in Close Touch

Buyers and sellers should have complete knowledge of transactionswithin the market. This is necessary so that they could be immediatelyaware of what is happening in any part of the market. This is possiblethrough the following means;

- Buyers and sellers to be present together in the same place.
- Modern methods of communication, such as telephones, postalservices, and internet facilities can easily bring buyers and sellersinto easy contact with one another in spite of the distancebetween them.

(d) Preferential Treatment

In a perfect market, there must not be any preferential treatment of favored customers or buyers and sellers.

(e) Free Entry and Exit into the Market

Buyers and sellers should be free to enter and leave the market. Thereshould be no unionism of any kind that will hinder or limit the extent ofentry and exit.

(f) Portability of Commodity

Commodity in a perfect market should be capable of being easilytransported from one part of the market to another.

Imperfect Markets

Any deviation from those conditions that make a perfect market makesthe market an imperfect one. Therefore, under imperfect markets thefollowing features are discernable;

(i) Non-homogeneous Commodity

The same type of a commodity may be branded under different namesor may be differentiated on the basis of further processing.

(ii) Entry and Exit

Entry into imperfect market is not free. A prospective seller must joinan existing association before he or she can be allowed to entry themarket while buyers may be compelled to buy from particular sellers.

For example, farmers' wives who bring the produce to their consumers in most markets have to sell first to some designatedmiddlemen who will eventually distribute to the retailers and thewholesalers before the final consumers can get the commodity to buy.

(iii) Preferential Treatment

Sellers in imperfect markets treat buyers differently since marketinformation is hoarded. Sellers charge differential prices depending onwho they are selling to.

Cases of Imperfect Markets

(a) Monopoly:

In a monopolistic market, two conditions must be fulfilled:

- There is only one seller selling the commodity or the few sellerscombine together to fix supply and price of the commodity.
- There is no substitute for the commodity.

(b) Oligopoly

In an oligopoly market;

- There are a few sellers and many buyers.
- It is called duopoly when there are just two sellers.
- Duopoly is thus a special case of oligopoly.

Volume of Trade

Markets can be classified on the basis of volume of trade and resourcebase of the operators. There are retail and wholesale markets.

a) Retail Market

- The sellers in this market are small traders.
- The sellers have small resource base hence they could onlyhandle small volume of the commodity they sell.
- The retailers sell directly to the final consumers.

b) Wholesale Market

- The wholesalers are big traders.

- They have huge resource base.
- They handle large volume of the commodity market. and perform

Commodity Traded

It is assumed that markets exist for the buying and selling ofcommodities.

(a) Forms of Commodities

Commodities can assume many meanings such as;

- Consumer goods or final products such as, tea, rice, beef, eggsand so on.
- Producer goods or intermediate products such as, raw materialssuch as coffee, cotton, sugarcane and so on.
- Production resources such as labour, capital, land and so on.

(b) Forms of Market

Using the various meanings of commodities we can therefore talk of thefollowing markets:

- Markets for final products: yam market, rice market and so on.
- Markets for Intermediate products: cocoa market, groundnutmarket, leather market and so on.
- Markets for Production resources: labour market, land market.
- Financial market such as; capital market for new securities,money market for short term loans, stock exchange for transfer ofsecurities and foreign exchange market for foreign exchange.

AGRICULTURAL MARKETING

Definition of Marketing

i. Agricultural marketing is the performance of all activitiesinvolved in the flow of goods and services from the point ofinitial production until they are in the hands of the ultimateconsumers.

ii. It is the process of satisfying human needs by bringing products topeople in the proper form, at the proper time and place they needthe products.

iii. Marketing however has several other meanings from theperspective of product, business firms and the society as a whole.

While the consumers are interested in getting what they want atthe lowest possible cost, the produces are interested in obtainingthe highest possible returns from the sale of their products. Thevarious firms engaged in doing the various marketing tasks areinterested in the profitability of their particular business operation. The overall ruler of marketing activities in a societyorganized in a private enterprise framework is the consumer. Theconsumer is king.

iv. Marketing is the process through which producers andconsumers are brought into contact formally or informally for theexchange of goods and services involving the performance ofmarketing activities. It is a management function that organizedand directs all business activities involved in assessing andconverting consumer purchasing power into effective demand fora specific product or service and in moving it to the finalconsumer or user so as to achieve the profit target or otherobjectives set by the producer.

Importance of Efficient Marketing

Any efficient marketing system should address the following:

i. Supply of food crops and livestock in the form, place, and timethe consumers want them.

ii. Location of where there are surpluses of produce and bringingthem to where there are shortages.

iii. Marketing is important for countries whose products are exportoriented since earnings from such exports are used to financeother development programmes.
iv. Marketing is an indicator of consumer preferences through theprices they are prepared to pay and this also affects theproduction decisions of farmers as they are likely to producecommodities that have high demand.
v. Efficient marketing system leads to increase in employment, risein the standard of living of the people and wealth of thecommunity.
vi. It also leads to increased incentives for government to provideinfrastructures such as roads, water, electricity, storage facilities.
vii. It leads to stimulation of research into techniques of foodpreservation, preparation and processing to meet the needs andtastes of the people.
viii. It ensures that supplies of goods that are seasonal becomeavailable throughout the year with little variation in prices.

Marketing of Agricultural Products
The marketing of agricultural products is very unique because of thespecial characteristics of most of the agricultural products. Thesecharacteristics include;
(a) Seasonality
Agricultural products are seasonal, whereas demand for them is fairlystable throughout the year. It is the function of marketing to ensure thatthe seasonal agricultural products are available throughout the year.
(b) Bulkiness
Agricultural products are very bulky due to their high waters content.
For them to attract high values they have to be processed to reduce thewater content (changing the form utility) before they are marketed.
Examples of bulky agricultural products are cabbage, potato, sugar cane tomato etc.
(c) Perishability
Agricultural products are highly perishable. Vegetables cannot stay formore than two days after being harvested before consumption exceptthey are frozen or dried. This characteristic increases the marketing costbecause they must get to the final consumer as quickly as possible. Theproblem is compounded by the inefficient storage facilities in thecountry due to incessant power failure to operate the available coldrooms and refrigeration.
(d) Production in the Hands of Peasant Farmers
Over 75 percent of agricultural productions in Kenya are handled bysmall scale farmers. This group of farmers operates small holding thatare scattered over appreciable long distances. The surpluses from these farms cannot easily be assembled because of the deplorable state of thefeeder roads and networks in the rural areas of the country. Also, thetransportation system is very expensive in the rural areas due to highcost of vehicle maintenance and state of the rural road networks.
(e) Processing Cost
Most agricultural products are not consumed in the state they areproduced. The processes of processing them into the form expected bythe consumers increase the marketing cost.

Marketing Functions
(a) Definition of Marketing Functions

- These are the activities performed by a marketing system inpresenting goods to consumers in the form, place and time they wantthe goods so that both producers and consumers gain.
- Marketing functions describe what happens or what can happen tothe product between the time of production and time of purchase bythe final consumers.
- They are the major specialized activity performed in accomplishingthe marketing process.

(b) The Major Marketing Functions

The three major marketing functions include:

a) Exchange functions
b) Physical functions
c) Facilitating functions

(i) Exchange Functions

These are activities involved in the transfer of title to good. It is at thisstage that price determination enters the study of marketing. Theactivities involved include;

- Seeking out the sources of supply.
- Buying the products and activities associated with purchase
- Assembling of products.
- Selling; activities associated with merchandizing such as physicaldisplay of goods, advertising to create demand.

(ii) Physical Functions

These are activities that involve handling, movement and physical change of thegoods to solve the problem of when, what and where utilities in marketing.Thefunctions are:

- Storage
- Transportation
- Processing.

(iii) Facilitating Functions

They are activities that make possible the smooth performance of theexchange and physical functions. They include;

- Standardization
- Financing
- Risk bearing
- Marketing intelligence.

Problems of Agricultural Marketing

Agricultural marketing is bedeviled by a lot of problems. Some of theproblems arise because of the basic characteristics and problems of Kenya's agriculture.

Some of the problems of agricultural marketing include;

- Supply of agricultural produce from scattered farms
- Lack of transport facilities
- Inefficient handling, packaging, and processing facilities
- Inadequate storage and warehousing facilities
- Lack of uniform weights and measures
- Adulteration of produce
- Instability in prices
- Lack of market information
- Inadequate research on marketing

AGRICULTURE IN KENYA' S ECONOMY

The major contributions ofproductive agriculture to a nation's economic growth could be summarized thus:

1) Food production-provide adequate food for an increasing population
2) Capital accumulation for industrial development - raw materials for a building industrial sector
3) Provision of employment for rural population, scientists andmiddlemen-the distributive section of the economy
4) Foreign exchange earnings for effective balance of trade at theinternational scene

Even though, emphasis has shifted from agriculture to other sectors of the economy such as tourism, manufacturing sector etc, agriculture still remains;

- ❖ The main stay of the Kenyan economy.
- ❖ Farming still remains the major source of employment of the bulk of the adult population since a large proportion of Kenyans live in rural areas as full time farmers where they cultivate bothfood and cash crop for home consumption and for cash. About80% of the labour force is still employed in agriculture.
- ❖ Kenyan Agriculture still provides enough food for thepopulation with little importation. Thus the fear of famine maynot materialize in Kenya if the current tempo in the agriculturalfood production is maintained.
- ❖ Agriculture was and is still the leading contributor to the GrossDomestic Product (GDP) of Kenya. Agriculture isstill the single major contributor to the GDP with more than25% ofthe GDP coming from the sector alone.